Masari (مساري)

A smart platform for the university student's journey

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1. Idea and problem

Masari ("my path") is a single platform that accompanies the student from the first day of college until graduation and employment. At the same time it serves faculty members and the college administration. It replaces a set of scattered tools, each of which solves only a small piece of the problem.

Problems the platform addresses:

- Students do not understand the academic regulations (bylaws), and the same questions reach Student Affairs every year.
- Timetables are prepared by hand and contain conflicts.
- Grading programming assignments takes a lot of time, and copying between students is hard to detect manually.
- Faculty members have no quick way to verify that an exam is balanced and covers the course.
- Buying textbooks and their access codes happens on paper at the library, with crowds, queues, and staff overhead.
- Students graduate without knowing what they lack for the job market, and cannot easily find a suitable internship.

2. Users and roles

The platform has four roles, each with separate permissions:

Role	Main use of the platform
Student	Ask regulation questions, view timetable, submit programming assignments and receive feedback, buy textbooks and codes, analyze CV and skill gaps, find and track internships, use AI study aids.
Faculty member	Auto-grade programming assignments and review plagiarism results, analyze exams before approval, request diagrams for exam questions, review AI-generated educational drawings before publishing.
Student Affairs	Dashboard for student questions and timetables, generate announcement posters.
Administration	Manage textbook codes, sales, and revenue; manage internship opportunities; oversee the platform.

3. The six modules

The platform consists of six modules, each with a primary user and a clear task.

Module	Serves	What it does	Core technology
Mu'eed معيد	Students and Student Affairs	A smart assistant that answers questions about the regulations in Arabic, a timetable generator that resolves conflicts, and a dashboard for Student Affairs.	RAG over Arabic documents + Constraint Satisfaction
CodeJudge	Students and faculty	Automatic grading of programming assignments with test cases, similarity (plagiarism) detection between submissions, and feedback to the student in Arabic.	Code analysis using AST
Mizan ميزان	Faculty	Analyzes an exam before it is approved: table of specifications (test blueprint), Bloom's taxonomy levels, and course coverage.	Question classification with NLP
Kitabi كتابي	Students and administration	Buy textbooks and their access codes online; the code and the book are delivered instantly to the student's account.	Payment gateway + automatic code distribution
CareerMap	Students	Analyzes the student's CV and compares it with jobs, identifies missing skills, and produces an ordered learning plan.	Semantic skill matching
Tadreeb تدريب	Students and administration	Connects students with suitable internship opportunities, tracks them, and evaluates them.	Semantic matching + tracking

4. Module details

4.1 Mu'eed: regulations assistant and timetables

- Chat assistant answering students' questions about the college regulations, in Arabic, grounded in the official Arabic documents (Retrieval-Augmented Generation).
- Timetable generator that produces schedules without conflicts using constraint satisfaction.
- Dashboard for Student Affairs.
- Holds the course catalog and course descriptions, which feed Mizan and Kitabi.

4.2 CodeJudge: programming assignment grading

- Automatic grading of submissions against test cases.
- Plagiarism detection that still works when variable names are changed, by comparing Abstract Syntax Trees (AST).

- Feedback to the student in Arabic.
- Sends each student's weak points to CareerMap.

4.3 Mizan: exam analysis

- Analyzes an exam before approval.
- Builds the table of specifications (blueprint).
- Classifies each question by Bloom's taxonomy level using NLP.
- Measures coverage of the course, using course descriptions received from Mu'eed.
- Planned as a light version first: Bloom classification + table of specifications + report.

4.4 Kitabi: textbooks and codes (newest module)

- The books for each course appear automatically according to the student's year (level) and division, linked to the courses in Mu'eed.
- Payment by card, Fawry, or mobile wallet.
- The administration uploads the codes once, and the system assigns exactly one code to each student after payment.
- Administration dashboard: who bought, revenue, remaining codes, and students who have not bought yet.
- An electronic receipt for every transaction.

4.5 CareerMap: career readiness

- Analyzes the student's CV and compares it with job postings.
- Identifies missing skills and produces an ordered learning plan.
- Uses semantic matching between skills and jobs.
- Receives weak points from CodeJudge, and sends required skills to Tadreeb.

4.6 Tadreeb: internships

- Matches students with suitable internship opportunities (semantic matching).
- Tracks each internship and records its evaluation.
- Receives the required skills from CareerMap.

5. The AI layer

On top of the six modules sits an AI layer that uses their shared data. Proposals that are candidates to be implemented first are marked **Candidate**.

5.1 Student assistance

Feature	What it does	Depends on
Study planner	A daily study plan built from the timetable and exam dates, adjusted automatically when the student falls behind.	Mu'eed
Question bank from the book	Review questions generated from the course textbook, distributed across Bloom's levels.	Kitabi + Mizan
Audio summary	Each chapter becomes a short audio file in colloquial (Egyptian) Arabic.	Kitabi
Early warning	Alerts the academic advisor when a student's grades drop or the student stops submitting.	CodeJudge
Elective course advisor	Suggests courses based on the student's level, interests, and career goals.	CareerMap

5.2 Image generation

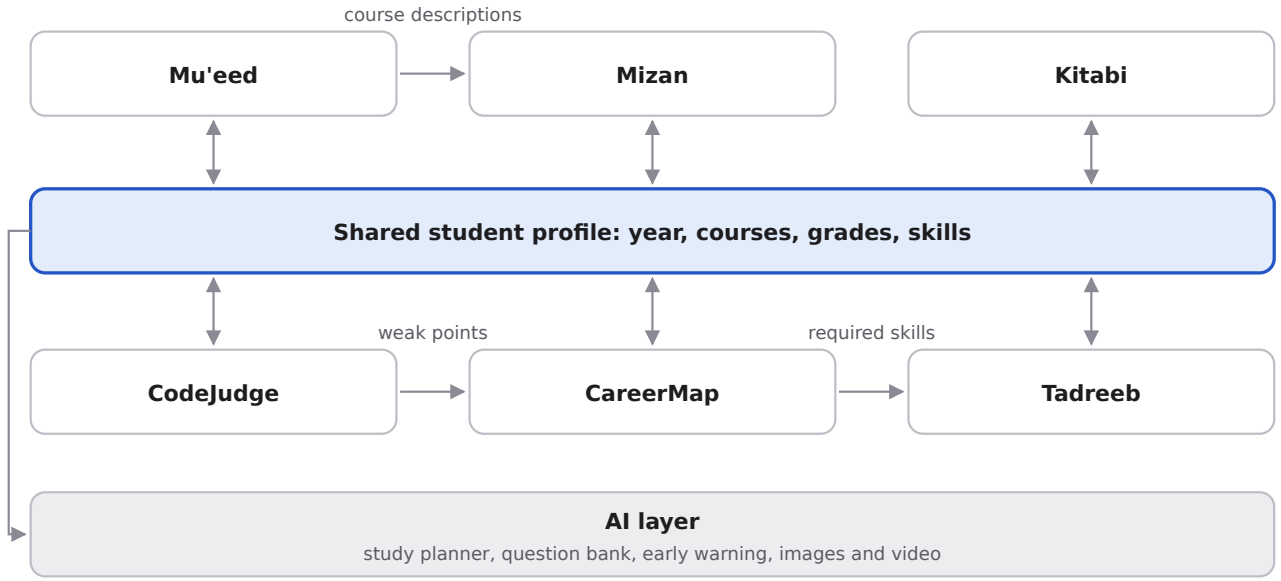
- **Concept drawing** **Candidate**: the student asks about a concept and gets an illustration or a flowchart.
- **Revision infographic**: a one-page summary of each chapter.
- **College posters** **Candidate**: Student Affairs writes an announcement and gets a ready poster; the same for internship announcements.
- **Exam diagrams**: a faculty member requests a drawing for a question (circuit, network, chart).

5.3 Video generation

- **Explain-in-a-minute** **Candidate**: part of a lecture becomes a short video made of animated slides with Arabic voice-over and written subtitles.
- **Welcome video for first-year students**: explains the platform and the regulations; regenerated every year from Mu'eed's data.
- **Course revision video** before the exam, covering the most important points.

6. Integration between modules

The modules share the student profile, so what happens in one feeds the others.



Module integration: six modules, one shared profile, one AI layer. Horizontal arrows are data passed directly between two modules.

Every module reads from and writes to the shared student profile. Three paths move data directly between modules:

- **Mu'eed → Mizan:** course descriptions.
- **CodeJudge → CareerMap:** the student's weak points.
- **CareerMap → Tadreeb:** required skills.

Kitabi takes the student's courses from the shared profile to display their books automatically.

7. Security

Security is a core part of the design, especially in Kitabi, because it handles payments and paid content.

- Every book code is bound to the student's account, so it cannot be sold or transferred to another student.
- Every downloaded PDF carries a watermark with the student's name and code, so the source of any leak can be identified.
- Codes are protected in the database, and every payment transaction is logged so any problem can be resolved easily.
- Separate permissions for each role: student, faculty member, Student Affairs, administration.
- The platform itself is tested against common vulnerabilities before delivery.

8. Technologies

Area	Use
RAG over Arabic documents	Answering regulation questions in Mu'eed
Constraint Satisfaction	Generating conflict-free timetables
AST-based code analysis	Detecting copying even when variable names are changed
Natural Language Processing	Classifying exam questions by Bloom's level, and matching skills to jobs
Payment gateway (sandbox / test mode)	Payments in Kitabi
Image, audio, and video generation	The AI layer

9. Why the project is strong

Masari's strength is that it is an integrated platform serving three groups with real technical depth, not a collection of side-by-side tools.

- **Integrated, not bundled:** the modules share the same data, so what happens in one feeds the others.
- **Serves three groups:** students, faculty members, and the administration.
- **Technical depth in every module:** RAG, scheduling algorithms, code analysis, Arabic language processing, and media generation.
- **Arabic and Egyptian:** built on Egyptian regulations, courses, and reality, not a translation of a foreign tool.
- **Feasible:** needs no approvals from external bodies and no large cost; the college's own data is enough for a pilot.
- **Scalable:** any other college can use it by changing only the regulations and courses.
- **Tangible impact:** less time spent by Student Affairs, conflict-free timetables, faster and fairer grading, balanced exams, and an end to textbook queues.

10. Implementation notes and scope

Six modules plus an AI layer is a heavy load for a single developer over about six months, so the project is built in phases with clear priorities.

Scope priorities

1. **Foundation:** accounts, permissions, and the shared database.
2. **The strongest modules for the demo:** Mu'eed, CodeJudge, and Kitabi.
3. **Complementary modules:** Mizan (light version: Bloom classification + table of specifications + report), CareerMap, and Tadreeb.
4. **AI layer:** start with the three candidate features, then the rest if time allows.

11. Risks and mitigations

Risk	Mitigation
Real payment gateways need a merchant account in the college's name, which is an external approval.	Use the payment gateway's test (sandbox) mode for the demo, and activate a real account later if the college adopts the platform.
Image generation models are weak at rendering Arabic text.	The model generates only the image; the Arabic text is added on top of it in code.
Generated educational drawings may contain scientific errors.	Generate drawing code (Mermaid or SVG) instead of an image, plus a faculty review button before publishing.
Generating a full video from text is expensive and slow.	Compose the video from images, audio, and subtitles with a tool such as FFmpeg.
Too many modules for a single developer.	Stick to the scope priorities above.

12. Open questions

Are the college's textbooks printed or PDF? If they are printed, Kitabi adds booking a pickup appointment with a QR code.

13. Glossary

Term	Meaning
Masari (مساري)	"My path": the name of the platform.
Mu'eed (معيد)	"Teaching assistant": the regulations and timetables module.
Mizan (ميزان)	"Scale / balance": the exam analysis module.
Kitabi (كتابي)	"My book": the textbooks and codes module.
Tadreeb (تدريب)	"Training / internship": the internships module.
Regulations (La'iha) (اللائحة)	The college's academic bylaws: rules for courses, grading, and progression.
Year / division / الفرقة) (الشعبة	The student's academic year (level) and specialization track.
Student Affairs شئون) (الطلاب	The college office handling student administration.
Fawry	A widely used Egyptian payment network.
Table of specifications	An exam blueprint mapping questions to course topics and cognitive levels.
Bloom's taxonomy	Levels of cognitive skill: remember, understand, apply, analyze, evaluate, create.
RAG	Retrieval-Augmented Generation: an AI model answers from retrieved source documents.
AST	Abstract Syntax Tree: a structural representation of code, independent of naming and formatting.
Constraint Satisfaction	An algorithmic approach that finds an assignment (e.g., a timetable) meeting all given constraints.